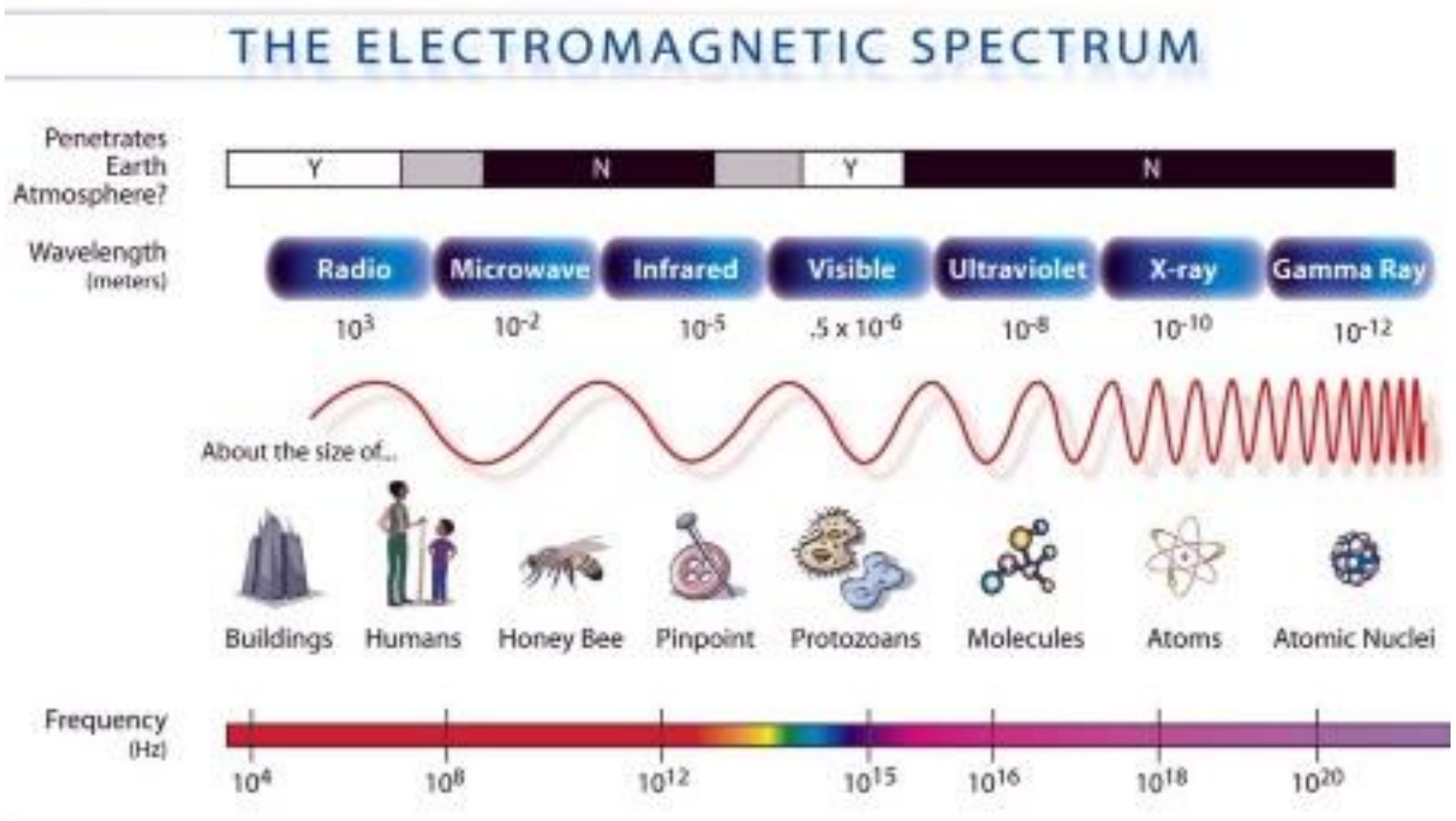
Molecular Spectroscopy and Spectral Regions



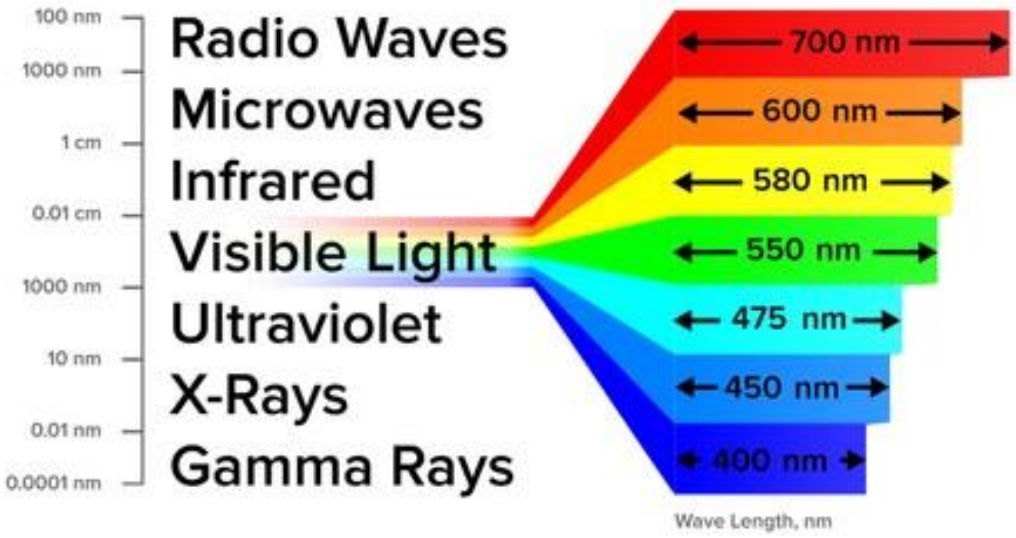
The study of the **interaction of electromagnetic radiation with matter** in all its forms, viz., solids, liquids/solutions and gases is called **Spectroscopy**.

The **radiations** are characterized by their **frequencies, wavenumbers, wavelengths and the energy associated with the electromagnetic wave**.

The **spectrum** in the domain of electromagnetic radiation is defined as **a series of radiant energy arranged in order of wavelength or frequency**.

Analysis of spectra gives rise to **information** regarding **energy levels and transition probabilities for atoms and molecules**.

Depending upon the energy of radiation (but not the intensity) the **spectroscopy has been classified into different branches**.



γ-rays – Nuclear transitions (**0.3 – 0.003 Å**) → oxidation states (environment of the nuclei)
x-rays – Electronic transitions (inner electrons) (**100-0.3 Å**) → core electron energies
Visible/UV – Electronic transitions (outer electrons) (**1 µm – 300 Å**) → electronic configuration
Infrared – Molecular vibration (**300-1 µm**) → stiffness of bonds
Microwaves – Electron spin, molecular rotation (**30-0.3 m**) → bond lengths and bond angles
Radio waves – Nuclear magnetic resonance, nuclear quadrupole resonance (**300-3 m**) → electronic structure near the nucleus

Nuclear Magnetic Resonance (NMR) Spectroscopy: $3x10^6$ – $3x10^{10}$ Hz; 10 m – 1 cm

NMR spectroscopy detects the **transitions between the nuclear states of the system in presence of applied magnetic field** and utilizes the **radio frequency** of the electromagnetic radiation.

Electron Spin Resonance (ESR) Spectroscopy: $3x10^{10}$ – $3x10^{12}$ Hz; 1 cm – 1 µm

Similar to NMR, ESR detects the **transitions between the various spin states of the unpaired electrons of the systems when placed in the external magnetic field** and utilizes the **microwave frequency** of the radiation.

Rotational Spectroscopy: $3x10^{10}$ – $3x10^{12}$ Hz; 1 cm – 1 µm

In rotational spectroscopy, the transitions are observed between the different rotational states of the molecules and utilizes the **microwave frequency** of the radiation.

Vibrational Spectroscopy: $3x10^{12}$ – $3x10^{14}$ Hz; 100 µm – 1 µm

Quantum states are evolved due to the **vibrational motion of the nuclei around their mean positions** and transitions occur between these states. This part of the spectrum falls in the **infrared radiation (infrared spectroscopy)**.

Raman Spectroscopy:

Raman spectroscopy also gives information about the **rotational and vibrational energy levels, not by absorbing, but by scattering of the monochrome beam of radiation**.

Electronic Spectroscopy: $3x10^{14}$ – $3x10^{16}$ Hz; 1 µm – 10 nm

Electronic spectroscopy involves the **transition between the electronic states of the system** and the radiation involved varies from visible to far UV region of the spectrum.

Basic Elements of an Absorption Spectrometer

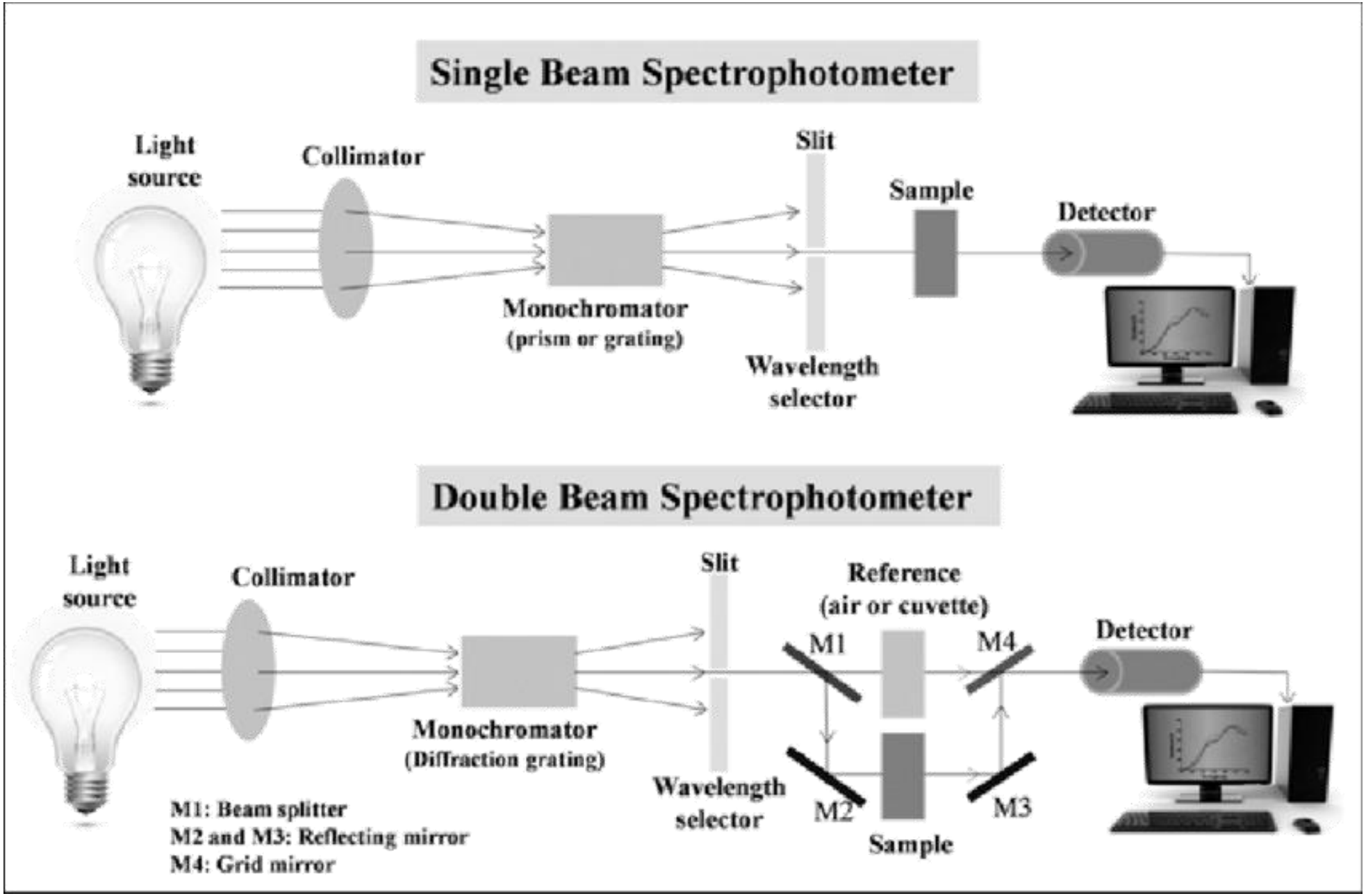
A beam of radiation falls on a sample → may induce transition between the ground and the excited states under resonance condition → The radiation falling on a sample may undergo absorption, transmission, scattering and reflection

$$I_0 = I_a + I_t + I_s + I_r$$

where I_a, I_t, I_s and I_r are the intensities of the respective phenomena. At the glass air interface, I_r is negligibly small and hence can be neglected. Thus, the equation becomes,

$$I_0 = I_a + I_t + I_s$$

The intensity of the absorbed radiation is measured as a function of energy with the help of an instrument called Absorption spectrometer. The curve between the absorbance versus wavelength is termed as absorption spectrum.

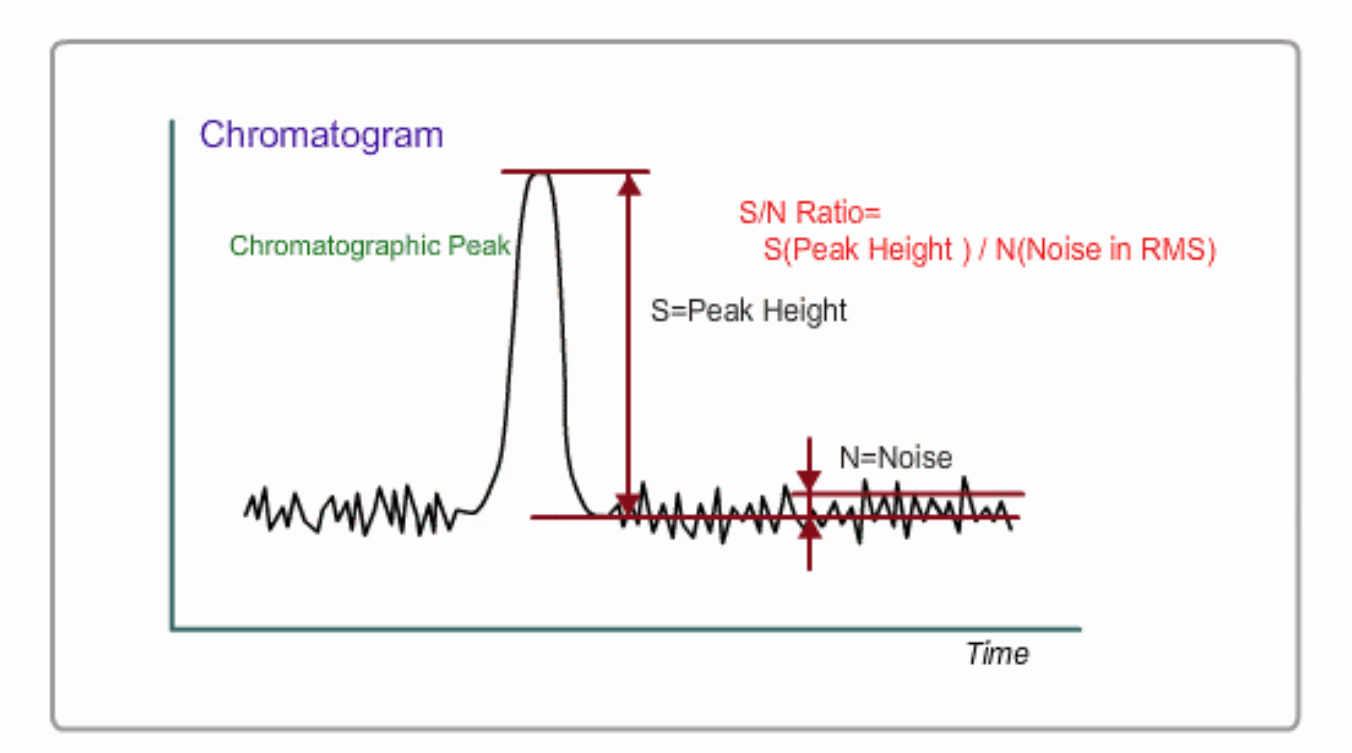

Single Beam Spectrophotometer
Light source
Collimator
Slit
Sample
Detector
Monochromator
(prism or grating)
Wavelength selector
Double Beam Spectrophotometer
Light source
Collimator
Slit
Reference
(air or cuvette)
M1
M4
Detector
Monochromator
(Diffraction grating)
Wavelength selector
M2
M3
Sample
M1: Beam splitter
M2 and M3: Reflecting mirror
M4: Grid mirror

Signal to Noise Ratio

The **spectrometer signal** produced by detector is **amplified** by electronic amplifiers.

Resulting **random fluctuations** due to electronic signals appear in the spectrum, which are called "**noise**".

The **sensitivity** of detection of the actual signal is maximum when **signal-to-noise ratio** is of the order of 3-4.



Resolving Power of Spectrometer and its Relation to Slit Width

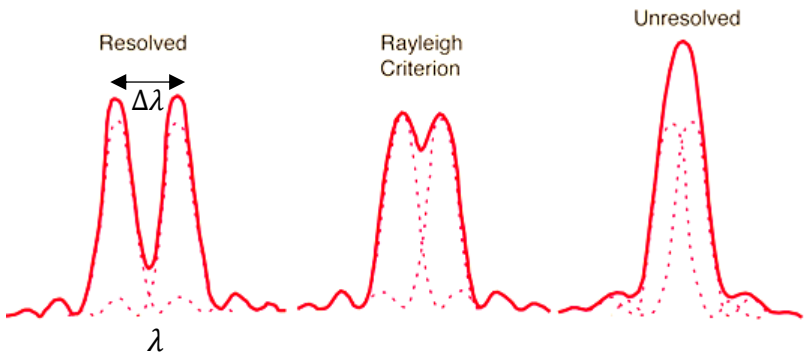
The efficiency of a spectrometer in separating the bands in a spectrum is called its <u>resolving power</u>.

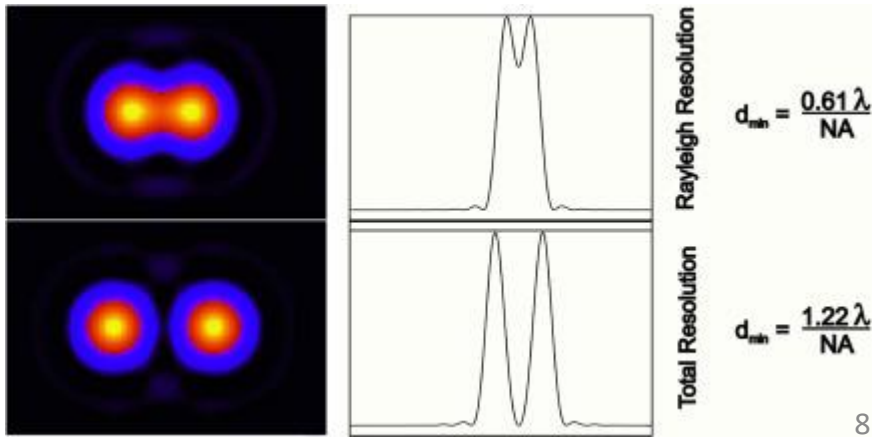
When two objects (or their images) are very near to each other, they may appear as one. The separation of such closed objects is termed as **resolution. The ability of an optical instrument to form distinctly separate images of two objects very close is called its <u>resolving power</u>.**

Resolving power is defined as

$$R_p = \frac{\lambda}{\Delta\lambda} = \frac{\bar{\nu}}{\Delta\bar{\nu}}$$

where λ or $\bar{\nu}$ is the average of two wavelengths or wave numbers and $\Delta\lambda$ or $\Delta\bar{\nu}$ is the smallest difference between two wavelengths or wave numbers that can be measured on a spectrometer.





$\frac{\lambda}{\Delta\lambda}$ is called "**theoretical resolving power**".

$$R_p = \frac{\lambda}{\Delta\lambda} = \frac{\bar{\nu}}{\Delta\bar{\nu}}$$

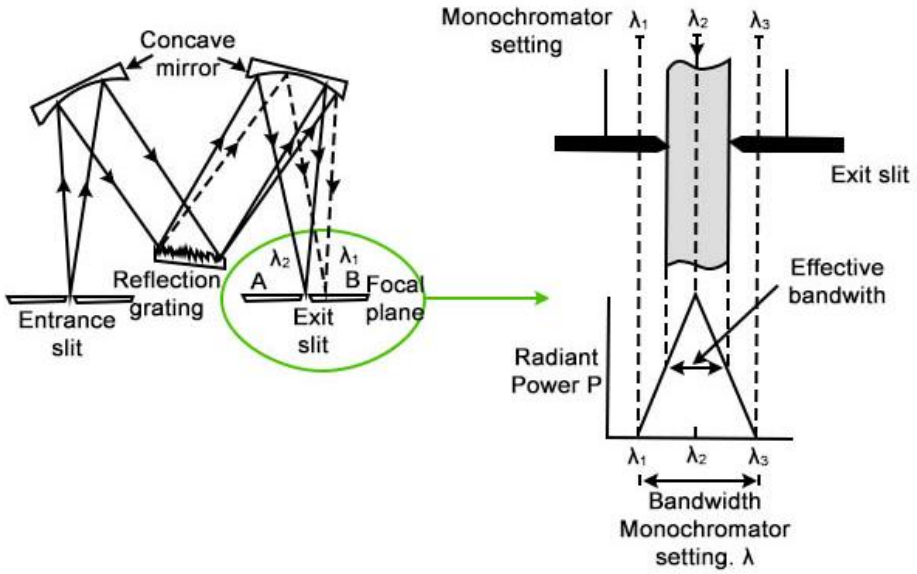
Molecular absorption does not take place at a single wavelength

but occurs over a large range of wavelengths that depends upon the width of the "**entry slits**"

The parameter, which **reduces the theoretical resolving power and width of slits**, is called the "**finite or <u>mechanical slit width</u>**".

The **small region of energy isolated by the exit slits** is called the "**spectral slit width**".

The **distribution of energy as a function of wavelength emitted from exit slits** is known as "**slit function**".



Resolution of instrument depends upon the spectral slit width and is proportional to the mechanical slit width.

Narrower slits result in better resolution.

The **energy received by the detector** is **proportional** to the **mechanical slit width**.

Thus, **narrowing of this width to improve resolution** has a **negative effect** on the **efficiency of the instrument**.

Narrow slit width allows less energy from beam to reach the detector.

The strength of the signal will decrease.

Thus, it is fixed at such a value where **proper signal-to-noise ratio is maintained**.

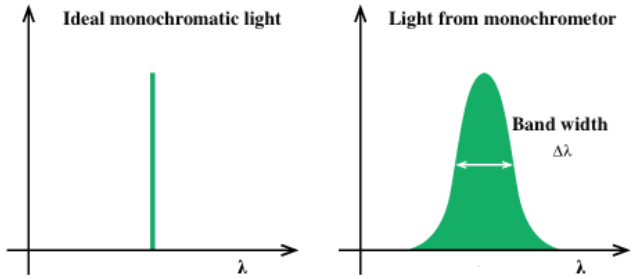
Since **large slit widths broaden the band**, so in any quantitative work **slit width should be less than the width of the band** under consideration.

When **slit width is equal to half the value of the bandwidth**, the **difference between the true and observed maximum extinction coefficient** is found to be of the order of **20 per cent**.

Width of the Spectral Lines and Factors Affecting it

The **inherent factor for line broadening** is the **Heisenberg uncertainty principle**.

Transition due to absorption of photons occurs between two defined energy levels → Therefore, the absorption bands shouldn't have any width at all → However, the observations are contrary to this hypothesis



The basic derivation of Schrödinger wave equation is based on Heisenberg uncertainty principle, which states that **if a system exists in an energy state for a limited amount of time Δt seconds, then the energy of that state will have uncertainty (ΔE) by the amount given by**

$$\Delta E \Delta t \geq \frac{h}{4\pi}$$

where ΔE is the uncertainty in the energy (i.e., energy spread) of the excited state, Δt is the mean time interval to increase the energy of the excited state.

Since the ground state has infinite lifetime ($\Delta t = \infty$) thus there is no uncertainty in its energy (ΔE=0).

However, the excited state is not well-defined. The energy (E) is not sharp but its spread over a certain range.

If the **uncertainty in the lifetime of the excited state is expressed in terms of its mean lifetime (τ)** and the **uncertainty in its energy is given by the width of the spectral line at half its maximum height, i.e., Γ_{nat}**, then

$$\Delta E\tau = h\Delta\nu\tau = \frac{h}{2\pi} \longrightarrow \Gamma_{nat} = \frac{1}{2\pi\tau}$$

Lifetime broadening

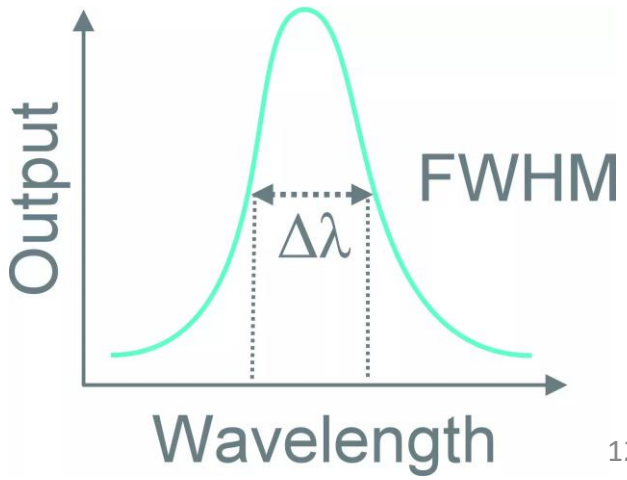
In other words, **the energy spread ΔE of the excited state manifests itself in the form of a band in the transition between the two states**.

The **width of the band** is called **natural band width ($\Delta\lambda$)** and is defined as **the width of the band at half of its height.**

The **broadening of spectral line in terms of wavelength** is expressed as

$$\nu = \frac{c}{\lambda} \longrightarrow d\nu = -\frac{cd\lambda}{\lambda^2} \longrightarrow |d\nu| = c\frac{|d\lambda|}{\lambda^2} \longrightarrow |d\lambda| = \frac{\lambda^2}{c}d\nu$$

Thus, $$\Delta\lambda = \frac{\lambda^2}{c}\Delta\nu$$



Phenomena which contribute to natural bandwidth

- **Pressure (or Collisional) Broadening**

Gaseous atoms (or molecules in liquid state) always coincide with each other, which perturb the energy levels resulting into broadening of the natural line width.

In each collision there is a certain probability that atom in an excited state will make a radiationless transition to the ground or a lower state so that the lifetime of the excited state will decrease.

- **Doppler Broadening**

In liquids and gases, the motion of the particles causes their absorption and emission frequencies to show a Doppler shift. **The apparent change in the frequency of radiation from an emitting atom in the excited state produced by its motion in the direction of the observer or the source of radiation is called Doppler effect**.

Since the motion is random in a given sample, shifts to both high and low frequencies occur and hence the spectral line is broadened.